EBU6505 – Reasoning and Agents
Block 2, Day 3 – Probabilistic Reasoning over Time

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Teaching Week 4

What have we covered this week?

- Day 1: Bayesian networks for uncertain reasoning
- Day 2: exact inference with enumeration and variable elimination
- elimination order and treewidth as cost intuition

Main idea: So far the model has been static. Today we ask what happens when the hidden state changes over time.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

What are we going to cover this week?

- Day 1: Bayesian networks
- Day 2: exact inference in Bayesian networks
- Day 3: probabilistic reasoning over time
- Day 4: planning with uncertainty
- Day 5: dynamic decision networks and partially observable Markov decision process synthesis

Main idea: This week moves from static uncertain belief to temporal belief, and then from belief to action.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

What are we going to cover today?

- why time makes probabilistic reasoning harder
- one simple running example: rain and umbrella
- filtering, prediction, smoothing, and most likely sequence
- hidden Markov models (HMMs), Kalman filters, and dynamic Bayesian networks (DBNs)
- exact versus approximate inference in temporal models

Main idea: The whole lecture answers one question: how do we track a hidden state when the world keeps changing?

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

Resources

- *Artificial Intelligence: A Modern Approach*, Russell and Norvig (4th ed.)
- AIMA 4e, Chapter 14: Probabilistic Reasoning over Time
- *Artificial Intelligence: Foundations of Computational Agents*, David Poole and Alan Mackworth (3rd ed.)
- Poole 3e, Chapter 9: Reasoning with Uncertainty
- Focus section for Poole: sequential probability models

Main idea: Both textbooks use the same core idea: the hidden state changes over time, but we only observe noisy evidence about it.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

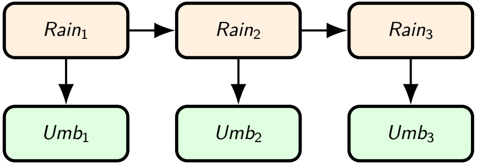
Learning Outcomes for Today

- explain why temporal reasoning is different from one-shot inference
- explain filtering, prediction, smoothing, and most likely sequence
- read the structure of a hidden Markov model (HMM) and a dynamic Bayesian network (DBN)
- explain when Kalman filters are more suitable than discrete hidden Markov models
- explain why approximate inference is often needed in larger temporal models

Main idea: If these five ideas are clear, students can follow the temporal-model lectures without getting buried in notation.

Warm-Up Question

Question: If you see umbrellas on three days in a row, what can you say about the rain on the third day?



What is hidden?

Whether it is raining each day.

What is observed?

Whether someone carries an umbrella each day.

Main idea: One observation can influence belief across several time steps.

Why Is Time a New Challenge?

Question: Why is one static Bayesian network not enough for a changing world?

Static case

one snapshot
one set of variables
one inference question

Temporal case

hidden state changes
new evidence keeps arriving
belief must be updated again

snapshot

$t_1 \rightarrow t_2 \rightarrow t_3$

Main idea: With time, inference becomes a repeated belief-update process.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

Our Running Example: Rain and Umbrella

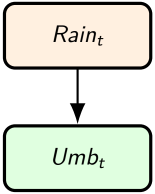
Question: What are the state and evidence variables in the simplest temporal model?

Hidden state

$Rain_t$
rain or no rain at time t

Evidence

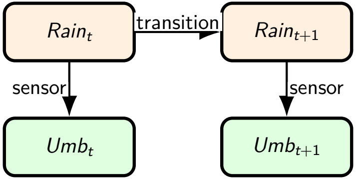
Umb_t
umbrella or no umbrella at time t



Main idea: We do not see the weather directly. We only see evidence about it.

How Do We Connect One Day to the Next?

Question: What dependencies appear when we move from one time step to many?



Transition model

How likely is tomorrow's state given to-day's state?

Sensor model

How likely is the observation, given the hidden state?

Main idea: A temporal model needs a state-change model and an observation model.

Four Core Temporal Questions

Question: What kinds of inference questions do we ask over time?

Filtering

What do we believe about the hidden state now?

Smoothing

What do we believe about an earlier state after later evidence?

Prediction

What do we believe about the hidden state later?

Most likely sequence

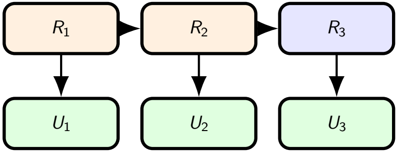
What is the single best full hidden-state path?

Main idea: These four questions organise the whole lecture.

Filtering

Question: What do we believe about the hidden state right now?

$$P(Rain_t \mid umb_1, \ldots, umb_t)$$



Simple meaning

Use all evidence up to today to estimate today's hidden state.

Example question

After three umbrella days, how likely is rain today?

Main idea: Filtering asks about the current hidden state.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

Filtering Uses Two Steps

Question: How do we update today's belief from yesterday's belief?

belief at $t-1$ → predict with transition → correct with evidence → belief at t

Predict step

Move the belief forward with the transition model.

Correct step

Adjust the belief using the observation seen today.

Main idea: Filtering repeats the same loop: predict first, then correct.

Prediction

Question: What do we believe about a future hidden state if no new evidence arrives?

$$P(Rain_{t+k} \mid umb_1, \ldots, umb_t)$$

belief now → $t+1$ → $t+2$ → $t+3$

Simple meaning

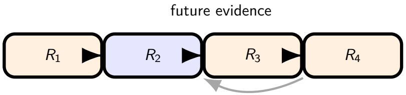
Start from the current belief and move it forward with the transition model.

Main idea: Prediction is a future-question.

Smoothing

Question: How can later evidence help us revise our belief about an earlier hidden state?

$$P(Rain_k \mid umb_1, \ldots, umb_T), \quad k < T$$



Simple meaning

Use evidence from before and after time k to improve the estimate for time k.

Main idea: Smoothing improves a past estimate using later evidence.

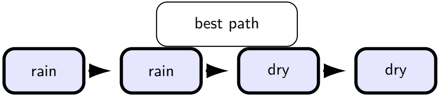
Most Likely Sequence

Question: What is the single best full hidden-state path across the whole timeline?

Different from smoothing

Smoothing estimates each time step separately.

Viterbi picks one best full sequence.



Example

Example path: rain $\rightarrow$ rain $\rightarrow$ dry $\rightarrow$ dry

Main idea: Viterbi chooses one best complete hidden-state path.

How Do the Four Questions Differ?

Question: How can we compare filtering, prediction, smoothing, and Viterbi quickly?

Question type	asks about	simple interpretation
Filtering	current state	what do we believe now?
Prediction	future state	what do we believe later?
Smoothing	past state	what do we now think happened earlier?
Most likely sequence	full path	what is the best complete hidden-state sequence?

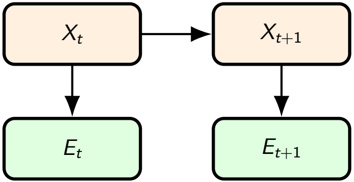
Main idea: All four are temporal-inference questions, but they focus on different parts of the timeline.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

What Is a Hidden Markov Model?

Question: What is the simplest standard temporal model for a hidden state and an observation?

- one hidden state at each time
- one observation at each time
- the next hidden state depends only on the current hidden state
- the observation depends only on the current hidden state



Main idea: A hidden Markov model (HMM) is the basic model for discrete hidden-state tracking over time.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

What Pieces Does an HMM Need?

Question: Which probability pieces define a full HMM?

Initial model

$P(X_1)$
where does the hidden state start?

Transition model

$P(X_t \mid X_{t-1})$
how does state change?

Sensor model

$P(E_t \mid X_t)$
how does evidence arise?

Main idea: An HMM is fully specified by one initial model, one transition model, and one sensor model.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

Small HMM Example: Rain and Umbrella

Question: What do transition and sensor probabilities look like in a real small example?

from / to	Rain	Dry
Rain	0.7	0.3
Dry	0.3	0.7

state	$P(Umb=t)$	$P(Umb=f)$
Rain	0.9	0.1
Dry	0.2	0.8

Transition idea

Weather tends to persist, but it can still change.

Sensor idea

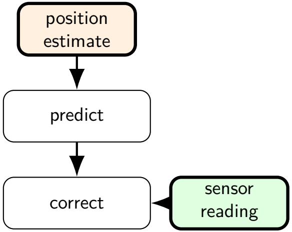
Umbrellas are strong clues for rain, but they are not perfect.

Main idea: The model is probabilistic because state change and observation generation are both uncertain.

Kalman Filter

Question: What if the hidden state is continuous instead of a small set of discrete labels?

- used for continuous quantities such as position, speed, or angle
- assumes linear dynamics and Gaussian noise
- repeatedly predicts and corrects, like filtering in an HMM



Example

robot position tracking, GPS fusion, sensor smoothing

Main idea: Kalman filtering is the continuous-state relative of HMM-style filtering.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

HMM or Kalman Filter?

Question: How do we know which temporal model is more suitable?

feature	HMM	Kalman filter
hidden state	discrete labels	continuous values
example	rain / dry	position / velocity
observations	categorized evidence	numeric readings
assumption	discrete transitions	linear Gaussian dynamics

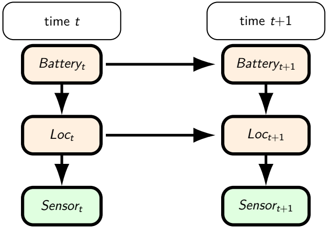
Main idea: Use HMMs for discrete states and Kalman filters for continuous states.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

What Is a Dynamic Bayesian Network?

Question: What if one time step contains several hidden and observed variables?

- a dynamic Bayesian network (DBN) generalises the HMM
- each time step can contain several variables
- arrows can exist within a time step and across time steps



Main idea: A dynamic Bayesian network (DBN) repeats a small Bayesian network across time.

DBN Example: Robot Tracking

Question: Why is a DBN more suitable than a simple HMM for many real systems?

Hidden variables

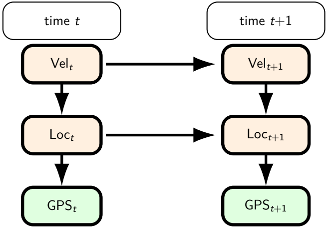
location, velocity, battery level

Observed variables

GPS, odometry, battery meter

Why useful?

Different hidden variables influence one another inside the same time step.



Main idea: A DBN lets us represent several related hidden and observed variables at once.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

Exact or Approximate Inference Over Time?

Question: Do temporal models always allow exact inference?

Exact inference

- forward algorithm
- forward-backward
- Viterbi

Approximate inference

- likelihood weighting
- particle filtering
- other sampling methods

Main idea: Small models can often use exact inference. Larger or more complex temporal models often need approximate inference.

Why Does Approximate Inference Become Useful?

Question: What makes exact temporal inference expensive?

more time steps or more variables → larger beliefs and larger factors → more time and more memory

Particle filtering is common when the state space is large, continuous, or hard to sum over exactly.

Main idea: Approximate methods are not second-best by design. They are often the only practical choice in realistic temporal systems.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

Temporal Reasoning in Intelligent Systems

Question: Where do we actually use these models outside textbook weather examples?

Robotics

track pose, velocity, and obstacles from noisy sensors

Energy systems

track load, faults, and risk from streaming sensors

Transport

track congestion and predict delays from live feeds

Healthcare monitoring

track patient condition from noisy measurements

Main idea: These models help when true state changes over time and evidence is noisy.

Applied Example: Robot Localization

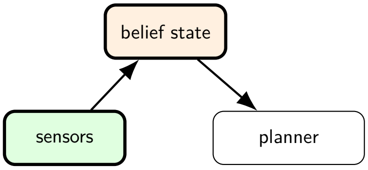
Question: What do filtering and prediction look like in a robot system?

System view

Hidden state: robot pose and velocity
Evidence: GPS, camera, lidar, odometry

Role

filtering stabilizes belief; prediction supports planning



Main idea: The planner should act on filtered belief, not on one noisy reading.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

Applied Example: Smart Transport Forecasting

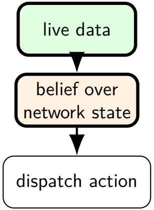
Question: What is the hidden state in a live transport monitoring system?

System view

Hidden state: demand, disruption severity, congestion level
Evidence: GPS traces, ticketing, incident reports

Role

prediction estimates delay risk before visible failure



Main idea: Prediction helps the system respond before disruption becomes obvious.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

Failure Modes in Temporal Reasoning

Question: What can go wrong if the temporal model or the evidence stream is poor?

Common problems

sensor drift
missing data
abrupt regime change
overconfident beliefs

Why it matters

a bad temporal belief can lead to bad planning, late alerts, or unsafe action selection

Main idea: The belief state is only as good as the model assumptions and the evidence stream behind it.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

Simple Mitigation Ideas

Question: How do we reduce temporal reasoning errors in practice?

Model-side actions

recalibration
drift monitoring
fallback models

Decision-side actions

be more conservative when uncertainty is high
wait for more evidence before irreversible actions

Main idea: Good systems track uncertainty, not just the most likely state estimate.

Where Are We Going Next?

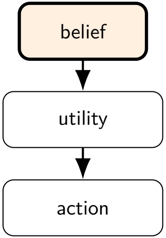
Question: What changes when we move from tracking belief to choosing actions?

Today

hidden-state tracking over time
infer what is happening now

Next lecture

combine belief, utility, and action
decide what to do



Main idea: Day 4 moves from belief tracking to action choice under uncertainty.

Key Takeaways

- temporal models track hidden state as new evidence arrives
- filtering, prediction, smoothing, and Viterbi answer different time-based questions
- HMMs are the basic discrete temporal model
- Kalman filters support continuous hidden states
- DBNs generalize temporal reasoning to several related variables per time step

Main idea: Students should now see temporal probabilistic reasoning as a belief-update process over time, not just a list of new algorithms.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458

In-Class Exercises

Think first → Work in pairs or small groups → Discuss and compare ideas

Use this time to work on today's in-class tasks and prepare one clear answer you can discuss.

Main idea: Use the exercise time to say clearly which temporal query or model fits each scenario and why.

Further Study

- Read Poole Chapter 9 and AIMA Chapter 14.
- Practice: Poole Ch.9, Exercises 9.13, 9.14, and 9.17.
- AIMA probabilistic-reasoning-over-time exercise page: 15.1, 15.2, 15.3, 15.13, and 15.20.
- Focus on filtering, smoothing, Viterbi-style sequence reasoning, and when continuous-state models are needed.

Refs: AIMA4e Ch.14 pp.461–499; Poole3e Ch.9 pp.375–458